Up to which $k_{\max}$ is linear galaxy bias enough?

A linear-bias fit to a one-loop truth on four bin layouts

November 2026

1 Summary

The question. The forecast fits the DESI galaxy clustering band powers C_ℓ^{gg} of thin redshift bins with linear galaxy bias, $\delta_g = b_1\,\delta$: the galaxy density contrast δ_g is the matter density contrast δ times one number b_1 per bin. This is exact on large scales. On smaller scales higher-order bias terms grow with the wavenumber k, so linear bias holds only below some scale cut $k_{\max}$. This note answers two questions about the four linear-bias priors of the correlated-prior note (`notes/correlated_bias_prior_v2`, section "The models").

1. Up to which $k_{\max}$ does a linear-bias fit to a one-loop truth stay unbiased, on each of four bin layouts?

2. Does the accuracy test of the linear-bias priors give the same verdict for Ω_m and A_s as for (w_0, w_a)?

Here $w(a) = w_0 + w_a(1-a)$ is the dark-energy equation of state, with a the scale factor; Ω_m is the matter density today in units of the critical density; and A_s is the amplitude of the primordial curvature power spectrum, sampled as $10^9 A_s$.

The answer.

1. Linear bias is safe only at the smallest scanned cut, $k_{\max} = 0.05\,h/\mathrm{Mpc}$, and for every prior and every truth only on the layout of equal comoving width. At $k_{\max} = 0.075\,h/\mathrm{Mpc}$, the cut of the correlated-prior note, a linear fit is biased on every layout and under every prior, and freeing the bias bin by bin does not cure it. The safe cut keeps 1–6% of the dark-energy figure of merit. The correlated-prior note therefore carries the higher-order bias at $0.075\,h/\mathrm{Mpc}$ with the one-loop model A.

2. Yes. Scored for Ω_m and A_s, the test excludes the growth-tied prior, finds the baseline prior honest only under its own scatter, and the pessimistic prior honest under both.

The setup. The data are the DESI band powers C_ℓ^{gg} alone, computed without the Limber approximation, for six tracer groups (BGS, LRG1–3, ELG1–2) on four bin layouts: bins of equal comoving width $\Delta d_{\mathrm{com}} = 62$ Mpc, and bins of fixed redshift width $\Delta z = 0.02$, 0.05 and 0.2.

Each one-loop truth is fitted with linear bias under four priors. With $D(z)$ the linear growth factor ($D(0) = 1$), $\bar{z}_i$ the mean redshift of bin i and B_g one amplitude per group g, they are: **growth-tied**, $b_i = B_g / D(\bar{z}_i)$; the **baseline prior**, a free amplitude and tilt per group plus a

residual of width $\sigma_g = 3\%$ for BGS and 1% for the other groups, correlated over $L = 0.2$ in redshift; the **pessimistic prior**, the same with $\sigma_g = 3\%$ everywhere and $L = 0.05$; and **free**, one bias per bin with a flat prior. The Gaussian covariance uses the growth-tied biases and is computed for 14,000 square degrees, rescaled to the survey's 12,300.

A fit's parameter shift is its Fisher linear response (Appendix A). The (w_0, w_a) shift is scored in units of the fit's marginalized error ellipse, and Ω_m and A_s each in units of its own marginalized error σ. A fit is unbiased when every shift is below $\sigma/3$, the threshold of the correlated-prior note's accuracy test.

The scale cut. $k_{\max}$ labels one cut per bin. Bin i is cut at $k_{\max}/D(\bar{z}_i)$: matter is less nonlinear at higher redshift, so the cut moves to higher k as $1/D$. The cut is the non-Limber ultraviolet-fraction criterion of Saraivanov et al. [1], which follows DeRose and Chen [2]. Each band power is split into $C_\ell^{\rm IR}$, from the power spectrum below the bin's cut wavenumber, and $C_\ell^{\rm UV}$, from the power above it. A band is kept while

$$\frac{|C_\ell^{\rm UV}|}{|C_\ell^{\rm UV}| + |C_\ell^{\rm IR}|} \;<\; \alpha = 0.05, \tag{1}$$

and the kept bands are intersected with the shipped mask, which caps every bin at $k = 0.1185\,\mathrm{Mpc}^{-1}$ with no growth scaling. The scan runs over $k_{\max} = 0.05, 0.075, \ldots, 0.25\,h/\mathrm{Mpc}$, and every $k_{\max}$ in this note is the $z = 0$ value.

The truths. The truth is model A of `notes/higher_order_bias`, computed with CosmoLike's one-loop engine. Its real-space galaxy power spectrum is

$$P_{gg}(k) \;=\; b_1^2\,P_{\rm mm}(k) \;+\; P_{\rm loop}(k;\, b_1, b_2, b_{s^2}) \;+\; \alpha_0\,k^2 P_{\rm mm}(k), \tag{2}$$

with $P_{\rm mm}$ the nonlinear matter power spectrum and $P_{\rm loop}$ the one-loop terms. These carry the quadratic bias b_2, the response of the galaxy density to δ^2, and the tidal bias b_{s^2}, its response to the squared tidal field. α_0 is the k^2 counterterm, in $h^{-2}\,\mathrm{Mpc}^2$, which absorbs the small-scale sensitivity of the loops; it multiplies $P_{\rm mm}$, CosmoLike's convention (DESI multiplies the linear spectrum). The *calibrated* truth keeps the fiducial b_1, puts b_2 on the Lazeyras et al. fit $b_2(b_1)$ [3] and $b_{s^2} = -\frac{4}{7}(b_1 - 1)$ on co-evolution [4], and sets the third-order bias $b_3 = 0$, $\alpha_0 = 0$ and no stochastic term. Four brackets offset every group's b_2 by ± 1 ("$b_2 \pm 1$", model A's prior width) or set $\alpha_0 = \pm 12.5\,h^{-2}\,\mathrm{Mpc}^2$, the counterterm prior width of the DESI 2024 full-shape analysis [5]. "Every truth" means all five.

The safe cut, layout by layout. Table 1 lists the largest safe $k_{\max}$: the largest scanned cut at which every shift stays below $\sigma/3$, at that cut and at every smaller cut that has a fit. Figure 1 draws the shifts against $k_{\max}$.

- $\Delta d_{\rm com} = 62$ Mpc: safe at 0.05 under every prior and for every truth; the largest shift at that cut is 0.14σ.
- $\Delta z = 0.02$: fails. At 0.05 the ultraviolet-fraction criterion keeps no band of any spectrum, so there is no fit; at 0.075 every prior fails.

- $\Delta z = 0.05$: safe at 0.05 for every truth under the growth-tied and baseline priors (largest shift 0.28σ). The pessimistic prior passes for the calibrated truth only, and fails the brackets $b_2 - 1$ and $\alpha_0 = -12.5$ with (w_0, w_a) shifts of 0.42 and 0.44. The free prior fails already for the calibrated truth (0.36).
- $\Delta z = 0.2$: safe at 0.05 for the calibrated truth under every prior, but no prior passes every truth: $b_2 - 1$ and $\alpha_0 = -12.5$ shift (w_0, w_a) by up to 0.49 (growth-tied) and 0.59 (the other priors). Its groups hold one or two bins each, so the free amplitude and tilt already free every bin, and the baseline, pessimistic and free priors coincide.

In every fitted case at 0.05 the largest shift comes from a negative bracket, $b_2 - 1$ or $\alpha_0 = -12.5$.

Layout	growth-tied	baseline	pessimistic	free
$\Delta d_{\rm com} = 62$ Mpc	0.05 / 0.05	0.05 / 0.05	0.05 / 0.05	0.05 / 0.05
$\Delta z = 0.02$	fails / fails	fails / fails	fails / fails	fails / fails
$\Delta z = 0.05$	0.05 / 0.05	0.05 / 0.05	0.05 / fails	fails / fails
$\Delta z = 0.2$	0.05 / fails	0.05 / fails	0.05 / fails	0.05 / fails

Table 1: The largest safe $k_{\rm max}$ [h/Mpc] of a linear-bias fit to the one-loop truth, as "calibrated truth / every truth". "fails": the smallest fitted cut already fails (0.075 on $\Delta z = 0.02$, where 0.05 keeps no band). $k_{\rm max}$ quoted at $z = 0$; each bin is cut at $k_{\rm max}/D(\bar{z}_i)$ by the UV-fraction criterion ($\alpha = 0.05$) within the shipped mask.

At $k_{\rm max} = 0.075\,h$/Mpc every prior is biased. For the calibrated truth the largest shift ranges from 0.40σ ($\Delta z = 0.2$, growth-tied) to 1.8σ ($\Delta d_{\rm com} = 62$ Mpc). On $\Delta d_{\rm com} = 62$ Mpc the four priors move Ω_m by 1.5–1.8σ, A_s by 1.0–1.3σ and (w_0, w_a) by 0.28–0.55; the step from 0.05 to 0.075 multiplies the shifts by factors of 6 to 53.

Why more bias freedom does not help. A linear bias multiplies a bin's power spectrum by the constant b_i^2 at every k. The loop terms and the counterterm of Eq. (2) have other shapes, so inside one bin the truth differs from every linear model by a scale dependence that no choice of b_i removes. The fit pushes that residual into the cosmological parameters, which do change the shape of the band powers. Freeing the biases changes which parameter takes the residual, not whether one does. On $\Delta d_{\rm com} = 62$ Mpc at 0.075, going from the growth-tied to the free prior lowers the Ω_m shift from 1.82 to 1.53 and raises the (w_0, w_a) shift from 0.28 to 0.55. On $\Delta z = 0.05$ the baseline prior moves the error from Ω_m (0.19, against 1.33 for growth-tied) to (w_0, w_a) (0.97, against 0.31).

The accuracy test for Ω_m and A_s. Figure 2 repeats the accuracy test of the correlated-prior note with linear-bias fits and linear truths at $k_{\rm max} = 0.075\,h$/Mpc. Each truth is a known bias drift (the $r < 19.5$ flux limit in BGS, a conserved galaxy population in the other groups) plus a residual drawn from the baseline or the pessimistic prior: 2000 draws, each fitted under every prior. The statistic is $\sqrt{\langle\chi^2\rangle/n_{\rm dof}}$, where χ^2 is the squared distance of the recovered parameters from the truth in units of the fit's marginalized covariance, $n_{\rm dof} = 2$ for (w_0, w_a) and 1 for Ω_m or A_s, and $\langle\cdot\rangle$ averages over the random truths and the data noise. It asks, for each parameter, whether the fit's quoted error bar honestly covers where the fit actually lands:

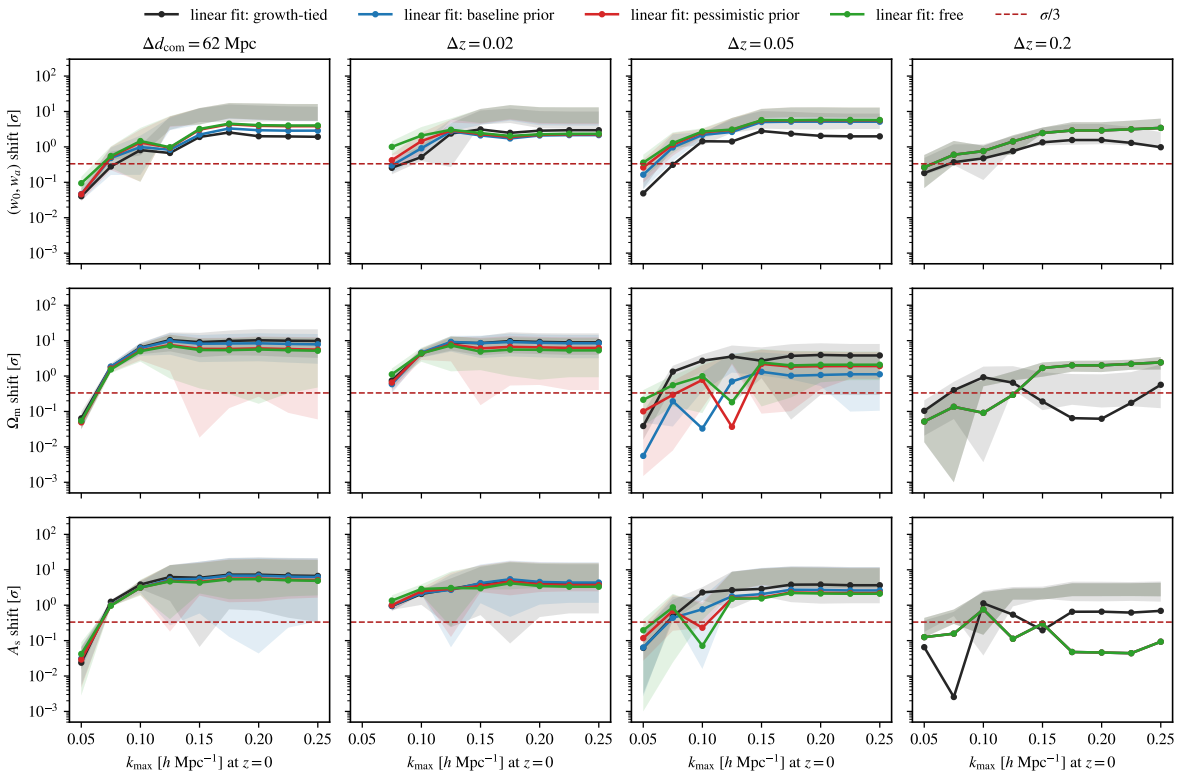


Figure 1: The shift of a linear-bias fit to the one-loop truth against the scale cut. Columns: the bin layouts. Rows: the (w_0, w_a), Ω_m and A_s shifts (Appendix A.3). Lines: the calibrated truth, one color per linear-bias prior. Bands: the range of the four brackets. Dashed line: $\sigma/3$. On $\Delta z = 0.02$ the cut at 0.05 keeps no band. On $\Delta z = 0.2$ the free line covers the baseline and pessimistic ones. $k_{\max}$ quoted at $z = 0$; each bin is cut at $k_{\max}/D(\bar{z}_i)$ by the UV-fraction criterion ($\alpha = 0.05$) within the shipped mask.

1 means it does, and above 1 it is the factor by which the error bar is too small. It scores the recovered parameters, not the χ^2 of the data: a biased fit can match the band powers well while its cosmology is wrong.

The verdict for Ω_m and A_s is the verdict for (w_0, w_a):

- **growth-tied** is excluded: the conserved-population drift alone shifts (w_0, w_a) by 5.0–11.6 and Ω_m by 0.48–6.2σ across the layouts;
- the **baseline prior** gives $\sqrt{\langle\chi^2\rangle/n_{\rm dof}} = 1$ under its own scatter (at most 1.02), but under pessimistic scatter its error bars are too small by up to 1.65 for (w_0, w_a), 2.0 for Ω_m and 1.48 for A_s ($\Delta z = 0.05$);
- the **pessimistic prior** is honest or conservative under both truths, with $\sqrt{\langle\chi^2\rangle/n_{\rm dof}} \leq 1.05$;
- **free** recovers every such truth exactly.

Every value is in `plots/ladder_bias_params/data.json`.

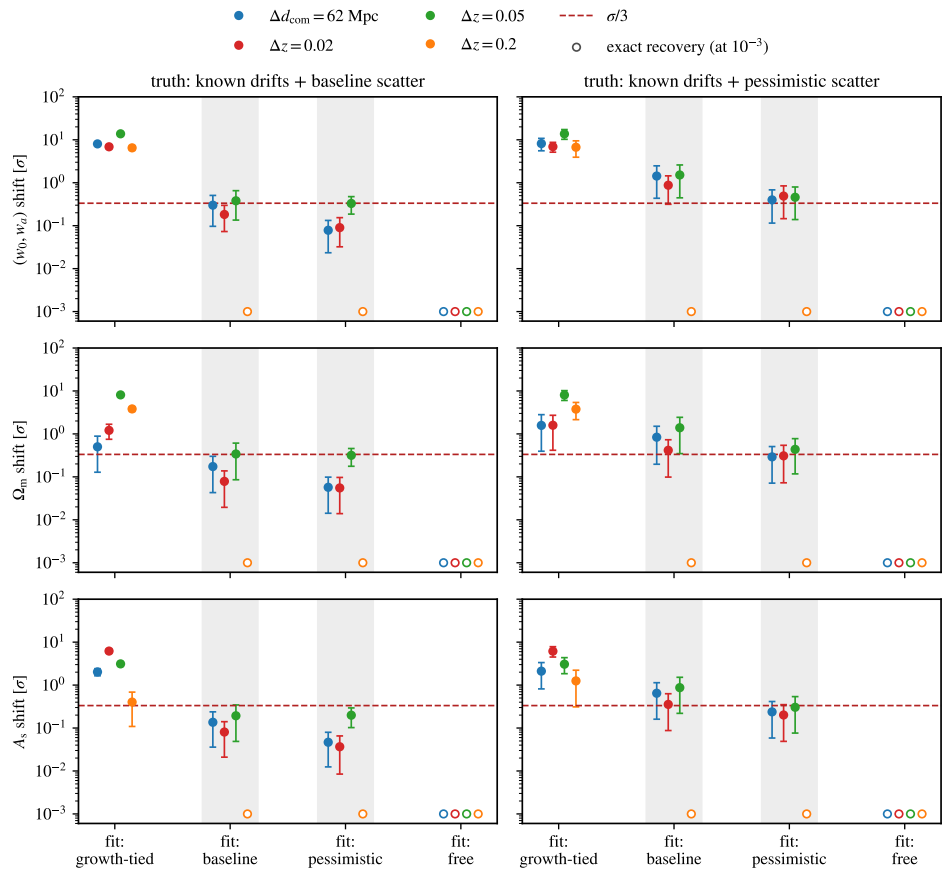


Figure 2: The accuracy test of the linear-bias priors, scored for (w_0, w_a), Ω_m and A_s. Truths: the growth-tied bias times $1 + \delta_i^{\rm drift} + \varepsilon_i$, with the known drift $\delta_i^{\rm drift}$ and a residual ε_i drawn from the baseline (left) or the pessimistic prior (right). Each truth is fitted under every prior (horizontal axis). Points: the mean shift over the truths; bars: 16th–84th percentiles; colors: the layouts; dashed line: $\sigma/3$; gray columns: the baseline and pessimistic priors. Where the fit's prior equals the truth's scatter, the fit's error covers the mean shift and $\sqrt{\langle \chi^2 \rangle / n_{\rm dof}} = 1$. Open markers at 10^{-3}: exact recovery. $k_{\rm max}$ quoted at $z = 0$; each bin is cut at $k_{\rm max}/D(\bar{z}_i)$ by the UV-fraction criterion ($\alpha = 0.05$) within the shipped mask.

The price of the safe cut. Figure 3 draws, for the three priors that pass the accuracy test, the dark-energy figure of merit against $k_{\rm max}$:

$$\mathrm{FoM}(w_0, w_a) \;=\; [\det \mathrm{Cov}(w_0, w_a)]^{-1/2},$$

the inverse area of the marginalized (w_0, w_a) ellipse. On $\Delta d_{\rm com} = 62$ Mpc the cut at 0.05 keeps 5% of the FoM at 0.075 under the baseline prior (0.26 against 5.1), 6% under the pessimistic prior and 1% under free. The loss is part of why the cut passes: it keeps fewer bands, on larger scales, where the one-loop terms are smaller and the data less precise.

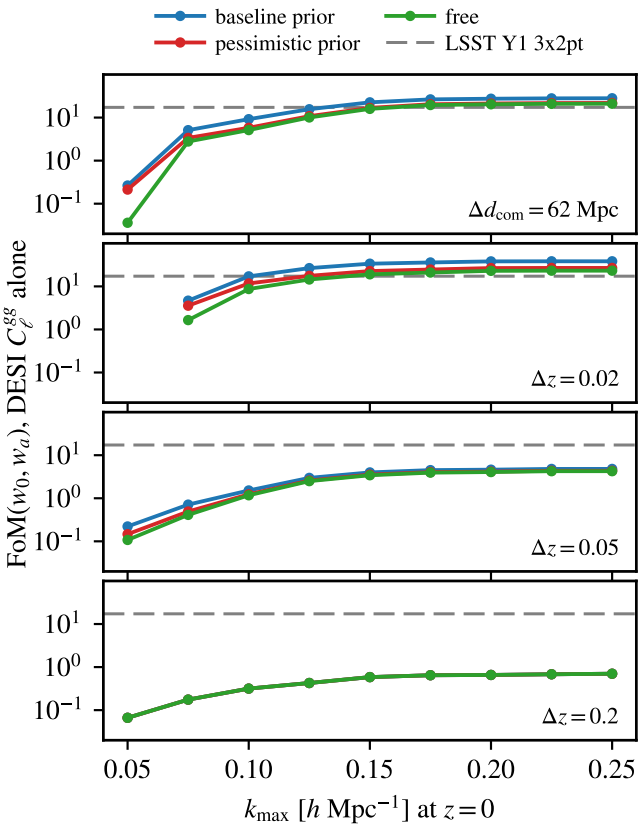


Figure 3: FoM(w_0, w_a) of a linear-bias fit to the DESI C_ℓ^{gg} alone against the scale cut, one panel per layout, for the priors that pass the accuracy test. Gray dashed line: the LSST Y1 3x2pt FoM. On $\Delta z = 0.02$ the cut at 0.05 keeps no band; on $\Delta z = 0.2$ the three priors coincide. $k_{\rm max}$ quoted at $z = 0$; each bin is cut at $k_{\rm max}/D(\bar{z}_i)$ by the UV-fraction criterion ($\alpha = 0.05$) within the shipped mask.

A Method

A.1 The truths

Each truth is a full model A data vector at the project's fiducial cosmology, with the linear biases at their fiducial values $b_{1,i} = B_g/D(\bar{z}_i)$. The calibrated truth uses the Lazeyras et al. separate-universe fit [3], $b_2(b_1) = 0.412 - 2.143\,b_1 + 0.929\,b_1^2 + 0.008\,b_1^3$, and the co-evolution value $b_{s^2} = -\frac{4}{7}(b_1 - 1)$ of a population formed with local bias and moved by gravity [4]. The brackets $b_2 \pm 1$ set every group's offset $\Delta_{2,g} = b_2 - b_2(b_1)$ to ± 1. The brackets $\alpha_0 = \pm 12.5\,h^{-2}\,\mathrm{Mpc}^2$ put that counterterm in every group: $\alpha_0 k^2 = 0.5$ at $k = 0.2\,h/\mathrm{Mpc}$, a 50% correction there [6]. CosmoLike's counterterm slot is $2\,b_1 b_K\,k^2 P_{\mathrm{mm}}$, with b_K its per-bin derivative-bias coefficient, so $\alpha_0 = 2\,b_1 b_K\,(c/H_0)^2$.

The vectors use the accuracy settings of the cached linear derivatives and the exact white-noise split of CosmoLike's low-rank engine. That engine factorizes $P_{gg} - P_\infty$, where P_∞ is the negative white-noise constant that the subtracted loop terms tend to at high k; P_∞ reaches C_ℓ through Limber, which is exact for a constant (`notes/higher_order_bias`, subsection "The white-noise limit of the renormalized loops").

A.2 The fit: linear response under each prior

Let m be the expected data vector, C its covariance on the kept entries (its inverse rescaled from 14,000 to 12,300 square degrees), and $\mathsf{D} = \partial m/\partial p$ the derivatives of the linear-bias model with respect to the fitted parameters p: the seven cosmological parameters ($10^9 A_s$, n_s, H_0, Ω_b, Ω_m, w_0, w_a, flat priors), the bias parameters of the prior, and the per-group velocity dispersions with their Gaussian priors. The fit's Fisher matrix is $F = \mathsf{D}^\mathsf{T}\mathsf{C}^{-1}\mathsf{D} + F_{\mathrm{priors}}$, where F_{priors} holds the velocity-dispersion priors and, for the baseline and pessimistic priors, the projected precision of the bias prior (the correlated-prior note, sections "The projected precision for two bins" and "The memory model").

If the data are the truth m_{truth} while the model at the fiducial predicts m_{lin}, then to first order the best fit moves by

$$\delta p \;=\; F^{-1}\,\mathsf{D}^\mathsf{T}\,\mathsf{C}^{-1}\big(m_{\mathrm{truth}} - m_{\mathrm{lin}}\big). \tag{3}$$

At the old best fit the gradient of the log posterior becomes $\mathsf{D}^\mathsf{T}\mathsf{C}^{-1}(m_{\mathrm{truth}} - m_{\mathrm{lin}})$ instead of zero, and F, the curvature, converts that gradient into a parameter displacement. The growth-tied prior uses the cached growth-tied derivatives, one bias column per group; the other priors use the per-bin derivatives, one column per bin. Both come from the linear notebook, `EXAMPLE_EVALUATE_FISHER.ipynb`.

A.3 The scores and the threshold

With δw the (w_0, w_a) entries of δp and C_w the (w_0, w_a) block of F^{-1}, the (w_0, w_a) shift is

$$\Delta_w \;=\; \sqrt{\delta w^\mathsf{T}\,\mathsf{C}_w^{-1}\,\delta w}, \tag{4}$$

the displacement of the dark-energy ellipse in units of its own size. For one parameter p the shift is $\Delta_p = |\delta p|/\sigma_p$, with $\sigma_p = \sqrt{(F^{-1})_{pp}}$. A shift of $\sigma/3$ added in quadrature to a 1σ error gives $\sqrt{1 + 1/9} = 1.05$: below the threshold the total error grows by at most about 5%.

A.4 Cuts without data and the safe $k_{\max}$

At each $k_{\max}$ the inverse covariance is rebuilt on the kept entries. The shipped mask's cap, $0.1185\,\mathrm{Mpc}^{-1}$, is about $0.176\,h/\mathrm{Mpc}$ at the fiducial $h = 0.6732$; at the largest cuts it, not Eq. (1), sets more and more of the kept bands, and the curves of Figures 1 and 3 flatten.

At small $k_{\max}$ some bins lose every band, and their bias columns hold only finite-difference noise. A parameter whose information (data and prior) at a cut is below 10^{-10} of its information on the full shipped mask therefore leaves the fit at that cut. A cut that leaves no data for the cosmological parameters gives no fit, as on $\Delta z = 0.02$ at $k_{\max} = 0.05\,h/\mathrm{Mpc}$. The safe $k_{\max}$ of Table 1 is the largest scanned cut at which every score of every truth considered is below 1/3, at that cut and at every smaller cut with a fit; a cut without a fit is skipped.

B Numerical checks

- **One fiducial vector.** The linear vector computed with the truths equals the cached reference of the linear notebook exactly on $\Delta d_{\mathrm{com}} = 62$ Mpc, $\Delta z = 0.02$ and 0.05. On $\Delta z = 0.2$ the largest relative difference, 4.5×10^{-3}, sits at a cross spectrum with nearly zero band powers; the median is 2×10^{-5}, and $\Delta\chi^2 = 2 \times 10^{-5}$ on the shipped mask.

- **Size of the one-loop sector.** The one-loop part of the calibrated truth is a median 1.9%, 1.6%, 2.3% and 1.9% of the linear band powers on $\Delta d_{\mathrm{com}} = 62$ Mpc, $\Delta z = 0.02$, 0.05 and 0.2.

- **The truths of the accuracy test.** The drifts and the 2000 random truths are drawn with the linear notebook's seed and in its order: they are the truths the correlated-prior note fits under model A. When the fit's prior equals the truth's scatter, the scatter and noise terms of $\langle\chi^2_w\rangle$ must sum to exactly 2; they do to 1.6×10^{-10}. The (w_0, w_a) value of $\sqrt{\langle\chi^2\rangle/n_{\mathrm{dof}}}$ is then within 0.03 of 1, the remainder being the drift.

- **The white-noise constant.** The exact projection of the one-loop white-noise constant leaves linear-bias vectors bitwise unchanged (`notes/higher_order_bias`).

C Limitations

- **Linear response.** The shifts are the first-order response of Eq. (3), not a nonlinear refit. Most verdicts rest on shifts well below the threshold or several times above it; a refit could move the cases near it, on $\Delta z = 0.05$ and 0.2 at 0.05 and growth-tied on $\Delta z = 0.2$ at 0.075 (0.40σ).

- **Theory truths, not mocks.** The truths are model A at and around its calibrations; a sample whose higher-order biases sit outside the brackets is not covered.

- **No stochastic term.** The truths carry no non-Poisson shot noise and set $b_3 = 0$. The effect of a constant noise power, which projects to an ℓ-independent C_ℓ, is not measured.

- **Real-space loops.** The truths add only the real-space one-loop terms of model A; the redshift-space one-loop terms of model B are absent from truth and fit alike.

- **Negative power in one bracket.** For BGS-like b_1 the $\alpha_0 = -12.5\,h^{-2}\,\mathrm{Mpc}^2$ truth makes P_{gg} negative above $k \approx 0.37\,h/\mathrm{Mpc}$, where the low-rank non-Limber correction floors it; this is immaterial at $k_{\max} \leq 0.075\,h/\mathrm{Mpc}$.

D Open issues

- **The covariance carries linear bias only.** Its signal is the linear-bias C_ℓ^{gg} with $b_i = B_g/D(\bar{z}_i)$; the one-loop terms of model A are left out. They are a median 1.6–2.3% of the band powers (Appendix B); their effect on the errors is not measured here.

- **The cut was built on the linear model.** Eq. (1) is evaluated on linear-bias band powers. The counterterm, with its k^2 weight, draws more of its signal from beyond the cut: on $\Delta z = 0.05$ at $0.075\,h/\mathrm{Mpc}$ the ultraviolet fraction of the α_0 template is a median 0.076 on the kept bands, above α, while for the full model A vector it is a median 0.017 and at most 0.052 (the correlated-prior note, subsection "Open: the counterterm's share beyond the cut").

References

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[4] S. Saito, T. Baldauf, Z. Vlah, U. Seljak, T. Okumura and P. McDonald, "Understanding higher-order nonlocal halo bias at large scales by combining the power spectrum with the bispectrum", 2014, arXiv:1405.1447.

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